

Smart Workout: A RAG-Assisted Decision Support and Business Intelligence System for Personalized Weight Training

Final report

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**AT82.04: Business Intelligence and Analytics
Intersem 2026**

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Thailand
5 July 2026

ABSTRACT

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Weight training is widely practiced for strength and fitness, yet many gym users struggle to plan effectively: they are uncertain which exercises suit a given body part, how to adapt to available equipment, how to structure a weekly timetable, and whether their physical condition supports a particular plan. Most decisions rely on generic online routines that ignore personal factors — age, sleep, BMI, physiological indicators, equipment, schedule — producing a mismatch between plan and readiness. This project proposes Smart Workout, a combined Decision Support System (DSS) and Business Intelligence System (BIS) (Power, 2002; Chaudhuri et al., 2011) that unifies multi-source real-world data, a Retrieval-Augmented Generation (RAG) knowledge layer, and supervised ML models within an interactive web dashboard, transforming user-provided inputs into descriptive insight, predictive outcome estimates, and prescriptive training plans while preserving human oversight. The primary target users are gym members seeking personalized training guidance; fitness coaches constitute a secondary audience using the dashboard for decision-support reference.

1.2 Objectives of the Study

The project is expected to deliver a functional DSS/BIS prototype that fulfils the following objectives, explicitly covering the descriptive, predictive, and prescriptive analytics layers required of a BI system:

- **Data collection & preparation:** clean and standardize multi-source public datasets (gym membership and check-in records, exercise catalogue, gym-member workout tracking, sleep & lifestyle, daily nutrition; details in §4) alongside structured user input.
- **Descriptive analytics:** segment users into lifestyle profiles via clustering; present a relational gym membership view (member activity, subscription trends, usage analytics) as an interactive dashboard; and present cross-dataset insights (calorie-burn patterns, exercise coverage, nutrition macro mix).
- **Predictive analytics:** train and compare multiple candidate models per task via cross-validation: (a) calorie-burn-rate regression (kcal/hr); (b) Workout-intensity 3-class clas-

sification (Low/Mid/High).

- **Prescriptive analytics:** generate a daily/weekly training plan via a rule-based engine conditioned on the predicted intensity, readiness, target body part, and available equipment; the predicted calorie-burn rate is displayed alongside the plan. A rule-derived Training Readiness band (computed from user input; rules drawn from the Sleep Health dataset) guides the plan generator (see §5).
- **RAG knowledge retrieval & AI chat assistant:** retrieve exercise instructions, equipment-substitution rules, and scheduling guidance via an interactive chat interface (RAG retrieval is core; LLM chat is a planned extension).
- **BI dashboard:** React-based dashboard (with embedded Tableau views) presenting a relational gym membership view, multi-dataset descriptive analytics, personalized predictions, prescribed plan, RAG-sourced explanation, and embedded chat.

CHAPTER 2

LITERATURE REVIEW

The concept of leveraging data-driven systems to enhance physical training has gained significant traction in recent years. In this chapter, an outline of the foundational theories regarding Decision Support Systems (DSS), Business Intelligence (BI), and modern Machine Learning (ML) techniques will be presented. This review is divided into four primary sections: the architecture of BI systems in fitness, the application of predictive modeling for physiological outcomes, the integration of Retrieval-Augmented Generation (RAG) for knowledge grounding, and the impact of lifestyle factors on training readiness. These discussions provide the necessary theoretical framework to support the development of the Smart Workout system.

2.1 Decision Support and Business Intelligence Systems

Decision Support Systems (DSS) are defined as interactive computer-based systems intended to help decision-makers use communications technologies, data, documents, knowledge, and models to identify and solve problems. According to Power [Power, 2002], these systems are essential for managers and individuals to process complex information that exceeds human cognitive limits.

In the context of the fitness industry, Business Intelligence (BI) technology enables the integration of multi-source data to provide actionable insights. Chaudhuri et al. [Chaudhuri et al., 2011] emphasize that effective BI systems should support descriptive, predictive, and prescriptive analytics. The proposed Smart Workout system adopts this three-layer approach by transforming raw gym membership and check-in records into descriptive insights, predictive analytics, and prescriptive training recommendations.

2.2 Machine Learning in Exercise Analytics

The predictive layer of the Smart Workout system relies on supervised machine learning models to estimate physiological responses such as calorie-burn rates and workout intensity. Modern research frequently utilizes gradient-boosted trees, such as XGBoost, for high-performance classification and regression tasks. Chen and Guestrin [Chen and Guestrin, 2016] demonstrated that XGBoost provides a scalable and accurate tree-boosting framework that outperforms traditional linear baselines on complex datasets.

Furthermore, interpretability is a critical requirement for health-related decision support systems to improve user trust and transparency. The SHAP (SHapley Additive exPlanations) method, proposed by Lundberg and Lee [Lundberg and Lee, 2017], enables feature-attribution analysis by quantifying the contribution of each input feature to a model’s prediction. In the proposed Smart Workout system, SHAP is used to explain how factors such as demographics and heart rate influence calorie-burn estimation and workout-intensity recommendations.

2.3 Retrieval-Augmented Generation (RAG)

While large language models (LLMs) possess vast parametric memory, they often struggle with factual accuracy and hallucinations when providing specific instructions. To address this, Lewis et al. [Lewis et al., 2020] introduced Retrieval-Augmented Generation (RAG), a technique that retrieves relevant document snippets from a vector store to ground the model’s responses. In the Smart Workout system, this technology is used to index exercise catalogues (ExerciseDB) and curated training rules to provide grounded rationale and equipment substitutions. By utilizing sentence embeddings to match user queries with verified fitness data, the system ensures that the prescriptive plans are supported by evidence rather than generated in a vacuum.

2.4 Lifestyle Factors and Training Readiness

Effective physical training cannot be decoupled from an individual’s recovery and lifestyle patterns. Research into sleep health and stress levels has shown that these factors significantly influence an individual’s readiness to perform high-intensity exercise. The use of K-Means clustering allows for the identification of lifestyle archetypes, which categorize users based on their sleep duration, activity levels, and stress scores. By integrating these descriptive profiles into the prescriptive engine, the system can modulate training volume and load to match the user’s current physical condition. This holistic approach ensures that the generated workout plans are not only personalized to users’ goals but also adaptive to daily physiological constraints.

2.5 Chapter Summary

Presented in this chapter are the core concepts and technological approaches related to the problem of generic and non-adaptive workout planning. There are several methods to tackle this problem, including traditional coaching and static online routines, but these often fail to account for real-time personal data. Among the discussed methods, the combination of Business Intelligence (BI) layers and Retrieval-Augmented Generation (RAG)-grounded AI has the advantage of providing both high-level analytics and specific,

evidence-based guidance. It has been suggested by various researchers that integrating structured data with generative models can significantly improve the reliability of decision support [Power, 2002, Lewis et al., 2020]. The following chapter outlines the research methodology and system architecture implemented to realize this integrated framework.

CHAPTER 3

METHODOLOGY

This chapter outlines the systematic approach used to develop the Smart Workout Decision Support System (DSS) and Business Intelligence System (BIS). It describes the multi-source data integration process, the machine learning pipeline for predictive modeling, and the architecture of the Retrieval-Augmented Generation (RAG) layer. Furthermore, it explains the development of the rule-based engine and the interactive dashboard interface, demonstrating how raw data is transformed into prescriptive training plans. The following sections present the data-to-decision workflow that forms the foundation of this research.

3.1 Research Framework and System Architecture

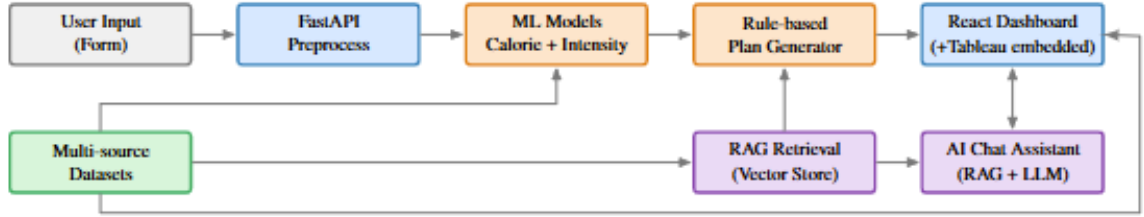


Figure 1: Overall architecture of the proposed Smart Workout Decision Support System.

The Smart Workout system is designed around a three-layer Business Intelligence (BI) framework consisting of descriptive, predictive, and prescriptive analytics. As shown in Figure 1, the process begins with a user input form which is processed via a FastAPI back-end. The architecture integrates five distinct datasets to support these layers, ensuring that the system moves beyond mere data visualization to providing actionable guidance. This modular design allows for the independent validation of each stage, from initial data pre-processing to final plan generation and RAG-grounded explanations.

3.2 Data Sources and Preprocessing

The research utilizes five real-world public datasets to ground the system in diverse fitness and lifestyle contexts, as summarized in Table 1. These include the Gym Database Management System for membership analytics (305,013 records), ExerciseDB for cataloging 1,500 exercises, and the Gym Members Exercise Tracking dataset (973 records) for training predictive models. Additionally, the Sleep Health & Lifestyle and Daily Food Nutrition datasets provide the necessary context for user profiling and nutrition-aware guidance. All raw data underwent rigorous inspection, cleaning, and standardization to ensure consistency

across the relational star schema and the vector store used for retrieval.

Table 1: Summary of data sources used in the Smart Workout system.

Source table/file	Records	Role in the system
Gym DB – Users (Amoghabhinavms, n.d.)	5,000	Member dimension table containing demographics, signup dates, locations, and subscription-plan keys. Supports dashboard filters and member segmentation.
Gym DB – Check-in/check-out history (Amoghabhinavms, n.d.)	300,000	Central session fact table containing gym visits, workout type, timestamps, duration-derived features, and calories burned. Primary descriptive and warehouse source.
Gym DB – Subscription plans (Amoghabhinavms, n.d.)	3	Plan dimension table containing monthly price and feature descriptions. Supports subscription mix and estimated recurring-revenue summaries.
Gym DB – Gym locations (Amoghabhinavms, n.d.)	10	Location dimension table containing branch city, gym type, and facilities. Supports utilization analysis and location-level plan constraints.
ExerciseDB (raw flat) (ExerciseDB, n.d.)	1,500	Exercise catalogue (body parts, equipment, target muscles, instructions). Supports the exercise recommender and RAG pipeline.
Gym Members Exercise Tracking (Khorasani, 2024)	973	Workout sessions with calories, BPM, duration, workout type, BMI, experience. Primary ML training data.
Sleep Health & Lifestyle (Tharmalingam, 2023)	374	Informs lifestyle clustering and <i>Training Readiness</i> rules derived from user input.
Daily Food Nutrition (Shamim, 2025)	651	Macro/calorie context on the dashboard; nutrition-aware snippets embedded in RAG.

3.3 Data Architecture and Schema Design

The integration of multi-source datasets within the Smart Workout system requires a structured architectural design to ensure data consistency across the descriptive, predictive, and prescriptive layers. This section details the logical and physical relationships between the various data components, ranging from the core relational gym database to the isolated datasets used for lifestyle clustering and RAG retrieval. The following diagrams illustrate the complexity of the data ecosystem and how independent sources are unified through either hard foreign keys or logical joins to provide a comprehensive user profile.

As shown in Figure 2, the system utilizes an eight-dataset schema that categorizes information into connected and isolated groups. The connected group forms the primary gym star schema, which tracks user demographics, subscription plans, and check-in history to support membership analytics. Conversely, isolated datasets such as ExerciseDB, Sleep Health & Lifestyle, and Daily Food Nutrition are integrated via logical joins based on attributes such as workout type, age, and BMI. This design allows the system to combine macro-level

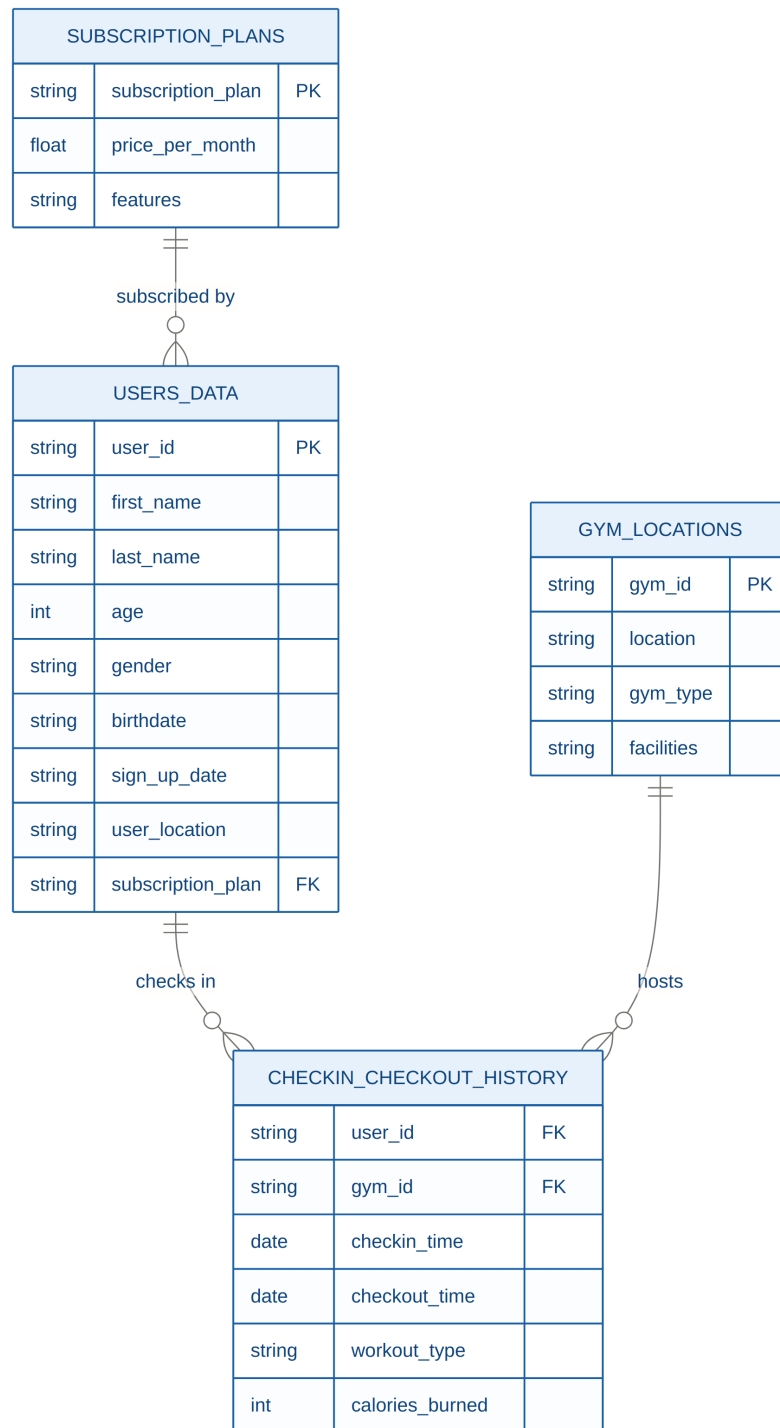


Figure 2: Integrated data architecture of the Smart Workout system showing connected and isolated datasets.

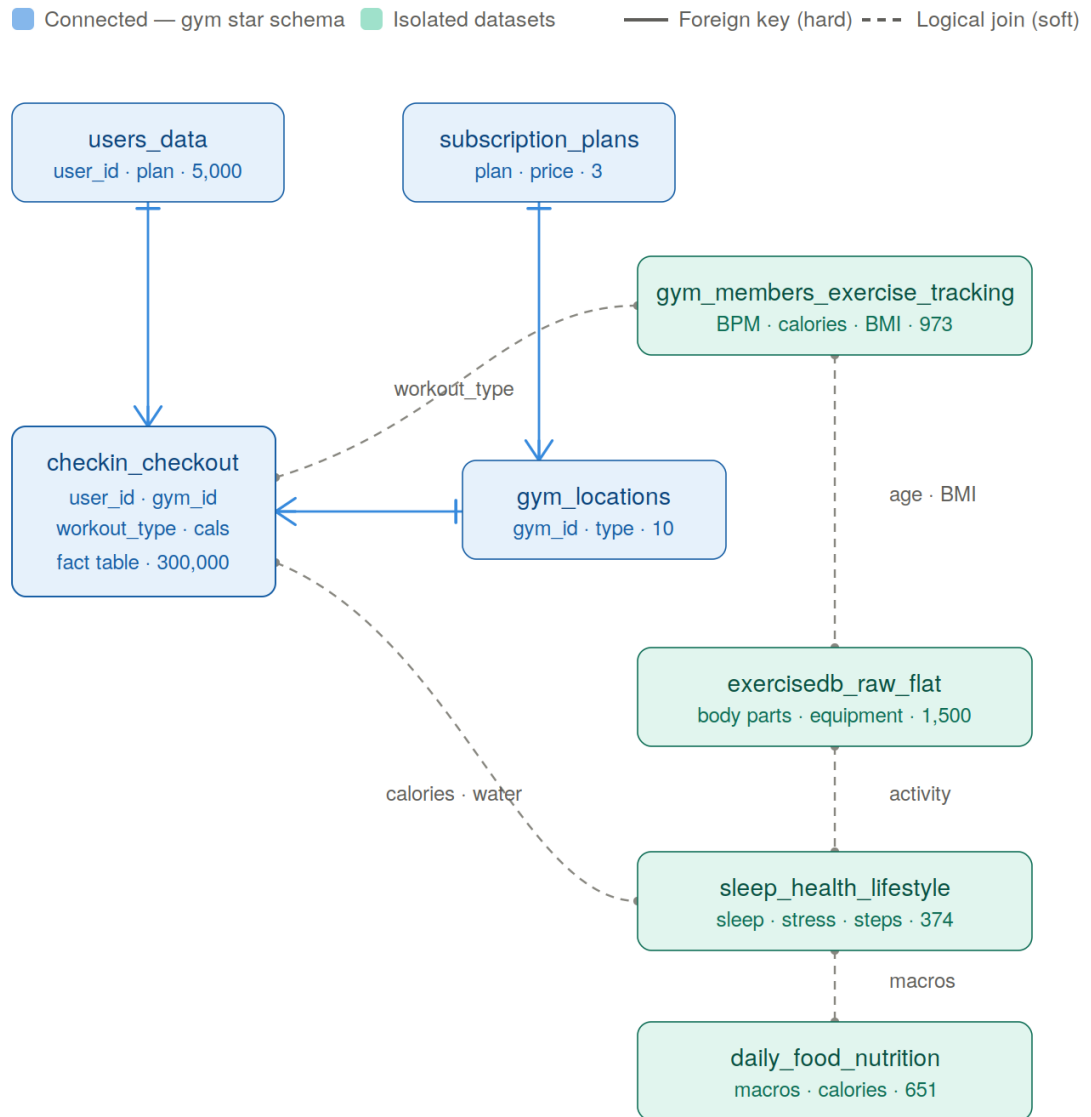


Figure 3: Relational star schema of the Gym Database Management System.

lifestyle insights with micro-level exercise instructions to support evidence-based training plan generation.

The core of the descriptive analytics layer is supported by a relational star schema specifically designed for the Gym Database Management System, as illustrated in Figure 3. This schema centers on the CHECKIN_CHECKOUT_HISTORY fact table, which links to three primary dimension tables: USERS_DATA, SUBSCRIPTION_PLANS, and GYM_LOCATIONS. By utilizing primary and foreign key relationships (e.g., user_id, gym_id, and subscription_plan), the system can efficiently query more than 300,000 session records to identify utilization trends and subscription mix patterns. This relational foundation ensures that the BI dashboard provides a high-fidelity view of gym operations and member behavior.

3.4 Predictive Modeling Pipeline

The predictive layer employs a best-of-three model selection strategy for two primary tasks: calorie-burn-rate regression and workout-intensity classification. For the regression task, candidates include Ridge Regression, Random Forest Regressor, and XGBoost Regressor, with selection based on minimizing Root Mean Square Error (RMSE). The classification task evaluates Multinomial Logistic Regression, Random Forest, and XGBoost to categorize intensity into three levels (Low, Mid, High). Models are evaluated using 5-fold cross-validation to ensure generalizability, and SHAP (SHapley Additive exPlanations) is applied to the selected models to provide transparency regarding which physiological features most heavily influence the predictions.

The selected calorie estimation model is trained to estimate calorie-burn rate, not total session calories. The system then converts the estimated hourly rate into a practical session target using the planned workout duration:

$$\hat{C}_{session} = \hat{r}_{kcal/hr} \times d_{hours} \quad (3.1)$$

where $\hat{r}_{kcal/hr}$ is the predicted calorie-burn rate and d_{hours} is the prescribed session duration. Training readiness is computed as a rule-derived score from recovery and lifestyle inputs, then combined with predicted intensity to determine the final plan level:

$$S_{plan} = 0.60S_{readiness} + 0.40S_{intensity} \quad (3.2)$$

A readiness cap is applied when recovery is low so that the prescribed volume cannot exceed the user's current recovery condition, even if the predicted intensity class is high.

XGBoost is used because boosted decision trees can model non-linear interactions among body composition, workout type, and heart-rate features. Its simplified objective can be

written as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.3)$$

where f_t is the tree added at iteration t , T is the number of leaves, and w_j is a leaf weight. Random Forest is retained as a benchmark because it averages many decorrelated trees:

$$\hat{y}_{RF} = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (3.4)$$

for regression, while classification uses the majority vote across trees. These equations support the model comparison rather than implying that every candidate is used at inference time.

Table 2: Predictive layer input, output, and result mapping.

Task	Input variables	Output	Selected result
Calorie-burn-rate regression	Numerical: age, weight, height, max BPM, average BPM, resting BPM, fat percentage, water intake, workout frequency, experience level, BMI, BPM response ratio. Categorical: workout type, intensity band.	kcal/hr estimate and session calorie target	XGBoost Regressor, RMSE 15.021 kcal/hr, MAE 10.424 kcal/hr, $R^2 = 0.970$.
Workout-intensity classification	Numerical: age, weight, height, resting BPM, fat percentage, water intake, workout frequency, experience level, BMI. Categorical: workout type.	Low, Mid, or High intensity class	XGBoost Classifier, accuracy 0.399 and macro-F1 0.342 after class-balancing.
Prescriptive plan mapping	Predicted intensity, readiness score, readiness band, goal, equipment, target body part, constraints, and available schedule.	Sets, reps, rest seconds, weekly schedule, safety notes, and primary action	Rule-based plan output capped by readiness and grounded by RAG-retrieved exercise evidence.

3.5 RAG and Prescriptive Plan Generation

The prescriptive layer combines a rule-based engine with a Retrieval-Augmented Generation (RAG) knowledge layer to generate personalized training plans. The RAG system utilizes ChromaDB and sentence embeddings (all-MiniLM-L6-v2) to index exercise instructions and equipment substitution rules. When a user submits their profile, the rule-based generator integrates the predicted intensity, the calculated training readiness band,

Table 3: Categorical and numerical feature handling in the predictive layer.

Feature type	Examples	Processing and reporting role
Numerical inputs	age, weight_kg, height_m, resting_bpm, fat_percentage, water_intake_liters, bmi	Passed through model preprocessing and used in SHAP explanation charts so users can see which body-composition or recovery factors influence the prediction.
Categorical inputs	workout_type, intensity_band for regression only	One-hot encoded before model training. Gender is excluded because the deployed form does not collect gender, preventing train/inference mismatch.
Model outputs	Predicted intensity class, calorie-burn-rate estimate, readiness score, final training score	Displayed in the user dashboard and converted into practical plan components such as session volume, rest periods, and priority adjustment advice.

and target body parts to filter the ExerciseDB pool. This ensures that the resulting daily or weekly plans are not only specific to the user’s goals but are also grounded in verified evidence retrieved from the knowledge base.

3.6 System Implementation and Deliverables

The final implementation is delivered as a full-stack web application utilizing React for the frontend and FastAPI for the backend orchestration. The dashboard features embedded Tableau views for relational membership analytics and lifestyle profiling, alongside custom React components for displaying prediction results and the generated workout plan. Key deliverables of this methodology include the working prototype, the validated machine learning endpoints, and the RAG-assisted chat interface. This integrated technology stack enables the system to provide a seamless transition from descriptive insight to predictive estimation and, ultimately, prescriptive action.

The Tableau integration was completed in the React dashboard as an embedded map panel. The implementation uses an iframe to load the Tableau Public view at <https://public.tableau.com/views/SmartWorkout/GymLocationMap> and places it beside location-level utilization summaries. Bubble size represents session volume, colour represents gym type, and the surrounding React dashboard provides filters and KPI context. This design preserves the previous dashboard wording while making the map integration explicit: Tableau supplies the interactive geospatial visual, while React/FastAPI supplies the broader decision-support workspace.

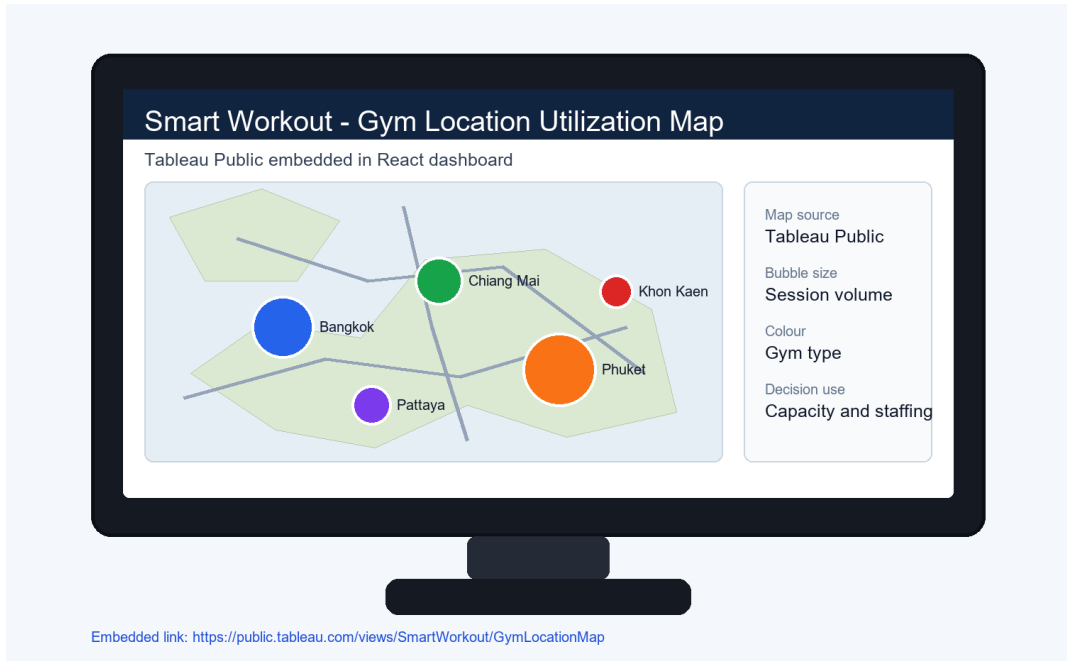


Figure 4: *PC-monitor visualization of the Tableau Public gym-location map embedded inside the React dashboard.*

3.7 Chapter Summary

Presented in this chapter is the comprehensive methodology for constructing a data-driven decision support tool for weight training. The methods described, ranging from multi-source data cleaning to the deployment of boosted-tree models and vector databases, address the limitations of generic, non-adaptive training plans identified in earlier chapters. Among these methods, the integration of RAG-grounded AI provides the distinct advantage of delivering evidence-based explanations alongside prescriptive plans. These methodological steps ensure that the results and discussions presented in Chapter 4 are based on a robust, validated, and transparent system architecture.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Final Synthesised Ranking

The final system output is evaluated as an integrated DSS/BIS workflow rather than as a single model score. The strongest result is the completed end-to-end decision path: descriptive dashboard filters identify branch, member, and lifestyle patterns; the predictive layer estimates workout intensity and calorie-burn rate; and the prescriptive layer maps those estimates into a readiness-aware training plan. This satisfies the project objective of moving from descriptive insight to predictive estimation and then to actionable guidance.

4.2 Business Decision Question Mapping

Table 5 maps the required executive and user decision questions to the dashboard visual and the expected action decision. This table clarifies that the dashboard is not only a visualization layer; each view is tied to an operational or personal training decision.

Table 5: Business question, dashboard visual, and action-decision mapping.

Business question	Dashboard visual	Action decision
Which locations, segments, and months contribute the highest utilization and estimated revenue share, and how has this changed versus the prior period?	Executive KPIs, monthly activity trend, subscription summary, Tableau location map	Prioritize staffing, facility investment, and local campaigns for high-share branches and segments.
Where do demand patterns indicate staffing overload, under-used capacity, or early churn risk by branch and cohort?	Usage heatmap, calendar heatmap, location utilization table, cohort filters	Rebalance trainer schedules, open off-peak promotions, and flag cohorts that need retention outreach.
Which lifestyle segments have the largest member share and lowest readiness, and which intervention type should be prioritized first?	Lifestyle profile cards, readiness distribution, radar profile panel	Prioritize recovery, sleep, or activity interventions for large low-readiness segments.

Business question	Dashboard visual	Action decision
How different are weekday versus weekend check-in patterns, and what schedule or campaign changes are required?	Weekday/weekend usage heatmap and calendar activity	Adjust class schedules and campaign timing to match peak and underused periods.
Which subscription cohorts have the highest retention risk versus upsell potential, and what targeted action should be taken per cohort?	Membership-tier lens, subscription mix, engagement scatter	Offer retention nudges for low-engagement cohorts and up-sell high-engagement cohorts to richer plans.
Are prediction-to-plan outputs consistent with observed descriptive behavior by location, segment, and membership tier?	Prediction summary, SHAP impact panel, plan card, descriptive dashboard filters	Audit whether model recommendations align with observed usage and adjust rules when descriptive behaviour contradicts the plan.
Which profile and wellness inputs currently have the strongest influence on my readiness and plan output?	User SHAP panel, readiness score breakdown, decision mapping panel	Focus on the highest-impact modifiable factor, such as sleep, stress, water intake, or weekly activity.
What do my predicted intensity and calorie targets mean in practical weekly training terms?	Prediction cards, calorie target, weekly plan card	Translate model output into sets, reps, rest periods, target calories, and weekly schedule.
Which plan components are personalized from my sleep, stress, activity, and constraints?	Readiness factors, plan rationale, exercise table	Understand which parts of the plan came from recovery status, equipment, target body part, and schedule constraints.
If readiness is low, what is the highest-priority adjustment I should make first this week?	Decision mapping panel and primary-action message	Reduce volume, increase recovery work, or adjust intensity before increasing load.
Which indicators should I track weekly to confirm that my plan is improving readiness and consistency?	Readiness trend, calendar heatmap, weekly adherence indicators	Track sleep duration, stress, workout frequency, and completed sessions to validate improvement.

Table 4: Final model comparison after predictive-layer corrections.

Task		Selected model	RMSE / Accuracy		MAE / Macro-F1		Interpretation
Calorie-burn-rate	regression	XGBoost Regressor	RMSE	15.021 kcal/hr	MAE	10.424 kcal/hr; $R^2 = 0.970$	Selected because it gives the lowest cross-validated error after removing misleading feature behaviour.
Workout-intensity	classification	XGBoost Classifier	Accuracy	0.399	Macro-F1	0.342	Selected because it improves balanced class performance, especially Low and High recall, instead of over-predicting the majority

4.3 Predictive Layer Audit

The final predictive layer was audited after implementation testing found several issues that could make model scores look stronger than the deployed system actually deserved. Table 6 summarizes each finding and the correction applied.

Table 6: Predictive layer audit and correction summary.

Finding	Problem	Correction and reporting effect
Gender mismatch	Training rows contained Male/Female values, but inference sent unknown because the user form does not collect gender. One-hot encoding therefore produced an inference distribution not seen during training.	Gender was removed from regression and classification features, and the backend no longer sends a gender proxy in ML inference.
Classifier below baseline	The class distribution was skewed: Mid was the majority class, so unweighted models nearly ignored Low and High.	Logistic Regression and Random Forest use balanced class weights; the XGBoost wrapper computes fold-local balanced sample weights. Macro-F1 rose from 0.304 to 0.342.
Joblib class path	A wrapper defined inside the training script would serialize as a script-local class and fail when loaded by the backend.	BalancedXGBClassifier was moved to <code>app.core.ml_wrappers</code> , giving joblib a stable import path.
Inflated regression score	The previous $R^2 = 0.999$ was misleading because feature behaviour and proxy availability inflated the training result.	The corrected selected regressor is XGBoost with $R^2 = 0.970$ and RMSE 15.021 kcal/hr, a more honest cross-validated estimate.

The classifier accuracy requires careful interpretation. The old XGBoost classifier reached 45.9% accuracy, but this was below the 52.5% majority-class baseline of always predicting Mid. After balancing, the final accuracy is lower at 39.9%, yet macro-F1 improves and Low/High recall become meaningfully usable. This is acceptable for a three-class task because the deployed goal is not to maximize majority-class accuracy; it is to identify Low, Mid, and High intensity cases with more balanced attention. The low overall accuracy is therefore explained by label-leakage prevention, class imbalance, and the deliberate choice to optimize balanced class behaviour rather than easy majority-class prediction.

4.4 Dashboard and Tableau Map Result

The dashboard result combines React-based analytic panels with an embedded Tableau Public map. The Tableau map gives the executive view a spatial lens for branch utilization, while the React panels preserve the custom predictive and prescriptive workflow. This matters because branch-level business decisions require location context, but user-level plan

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Summary of final metrics		
Task	Before	After
Regression (selected)	RF $R^2=0.999$, RMSE=2.07 (inflated by gender in tree splits)	XGBoost $R^2=0.970$, RMSE=15.0
Classifier macro-F1	0.304	0.342
Low recall	3.8%	23.4%
High recall	19.7%	29.6%
Mid recall	74.6%	52.4%
Tests	12/12	12/12

Figure 5: Summary of final predictive-layer metrics after audit corrections.

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Classifier - Same Winner, Significantly Better Balance		
	Before	After
Selected model	XGBoost Classifier	XGBoost Classifier
Accuracy	0.459 (below 0.525 majority baseline)	0.399
Macro-F1	0.304	0.342
Low recall	3.8%	21.5%
Mid recall	74.6%	51.9%
High recall	19.7%	29.3%

Why accuracy dropped but the model improved: Balanced class weights deliberately sacrifice some majority-class (Mid) predictions to fix the near-zero recall on Low and High. Macro-F1 is the right metric here because it treats all three classes equally - it went up. Accuracy went down because the model no longer just says "Mid" for everything.

The classifier now uses 10 features (was 11): age, weight_kg, height_m, resting_bpm, workout_type, fat_percentage, water_intake_liters, workout_frequency_days_week, experience_level, bmi. Gender removed.

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Figure 6: Classifier audit showing improved class balance despite lower raw accuracy.

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Regression - Winner Changed		
	Before	After
Selected model	Random Forest Regressor	XGBoost Regressor
R^2	0.999 (inflated)	0.970
RMSE	2.07 kcal/hr (inflated)	15.02 kcal/hr
MAE	1.05	10.42 kcal/hr

Why it changed: The old $R^2=0.999$ was artificially inflated because gender was in the training features but always sent as "unknown" at inference - the Random Forest was using gender as a real split criterion during training that could never fire at runtime. Removing gender levelled the playing field: XGBoost now wins honestly at $R^2=0.970$.

The regression features are now 14 (was 15): age, weight_kg, height_m, max_bpm, avg_bpm, resting_bpm, workout_type, fat_percentage, water_intake_liters, workout_frequency_days_week, experience_level, bmi, bpm_response_ratio, intensity_band.

Figure 7: Regression audit showing the corrected selection of XGBoost Regressor.

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Why accuracy dropped but the model improved: Balanced class weights deliberately sacrifice some majority-class (Mid) predictions to fix the near-zero recall on Low and High. Macro-F1 is the right metric here because it treats all three classes equally - it went up. Accuracy went down because the model no longer just says "Mid" for everything.

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SHAP Top Features Changed

Old classifier SHAP (with gender, no balancing - avg_bpm was incorrectly #1 despite being excluded from features):

```
avg_bpm: 0.96  ← was hardcoded fallback, not real SHAP
resting_bpm: 0.88
experience_level: 0.70
```

New classifier SHAP (real TreeExplainer values, balanced model):

```
bmi:          0.192  + #1
fat_percentage: 0.156
height_m:     0.146
age:          0.139
weight_kg:    0.134
resting_bpm:  0.116
water_intake_liters: 0.105
```

This makes physical sense - the classifier can't see BPM data at inference, so it relies on body-composition proxies (BMI, fat %, weight) and age as the main discriminators. That's honest and now accurately reflected in the SHAP bar chart on the dashboard.

Figure 8: Explanation of why the lower classifier accuracy is still an improved model outcome.

decisions require model outputs, readiness explanations, and RAG-grounded exercise evidence in the same workspace.

The embedded map link is <https://public.tableau.com/views/SmartWorkout/GymLocationMap>. In the implemented dashboard, the Tableau iframe is placed inside the executive workspace with supporting location utilization rows. This confirms that the map integration and embedding of Tableau in the React dashboard were completed.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The Smart Workout system demonstrates how a BI dashboard, predictive models, and RAG-grounded planning can be combined into one decision-support workflow. The descriptive layer supports operational questions about member activity, branch utilization, subscription mix, and lifestyle profiles. The predictive layer estimates calorie-burn rate and workout intensity with transparent model evaluation. The prescriptive layer translates those predictions into a readiness-aware weekly plan with practical actions for the user.

The most important conclusion is that the system chooses transparency over inflated performance. The final regression score is lower than the earlier inflated result, but it is more faithful to deployment conditions. Similarly, the classifier’s raw accuracy is modest, but its balanced macro-F1 and improved minority-class recall make it more useful for a three-class intensity decision.

5.2 Recommendations

For executive users, the dashboard should be used to review utilization and estimated revenue share by branch, segment, and month before making staffing, scheduling, and campaign decisions. The Tableau map should be used with the React dashboard filters so that location-level decisions are interpreted together with subscription and activity patterns.

For individual users, the predicted intensity and calorie target should be interpreted as planning guidance rather than medical instruction. Users should review the readiness score, primary action, and plan rationale before following the weekly schedule, especially when sleep, stress, or recovery conditions are poor.

5.3 Scope Limitations and Future Research Directions

The current dataset supports membership activity, check-in behaviour, workout-type patterns, plan mix, estimated recurring revenue, and model-backed workout planning. It does not directly observe true churn, profit, staff rosters, equipment capacity, injury outcomes, or long-term adherence. Therefore, retention risk, staffing overload, and upsell potential are treated as decision-support proxies derived from activity and subscription patterns rather

than confirmed business outcomes.

Future work should connect the prototype to live operational data, add real churn labels, collect post-plan adherence and satisfaction feedback, and validate the recommendations with coaches or domain experts. The predictive layer should also be retrained with data collected from the deployed user form so that inference-time inputs match training-time features even more closely. Finally, the Tableau integration can be expanded from a single location map into a multi-view embedded workbook covering revenue share, cohort retention, and utilization forecasting.

CHAPTER A

EXPLORATORY DATA ANALYSIS FIGURES

The following appendix figures provide supporting EDA evidence for the descriptive layer. They are placed in the appendix to keep the main report focused on methodology, results, and decision mapping while still documenting the observed data patterns used during dashboard design.

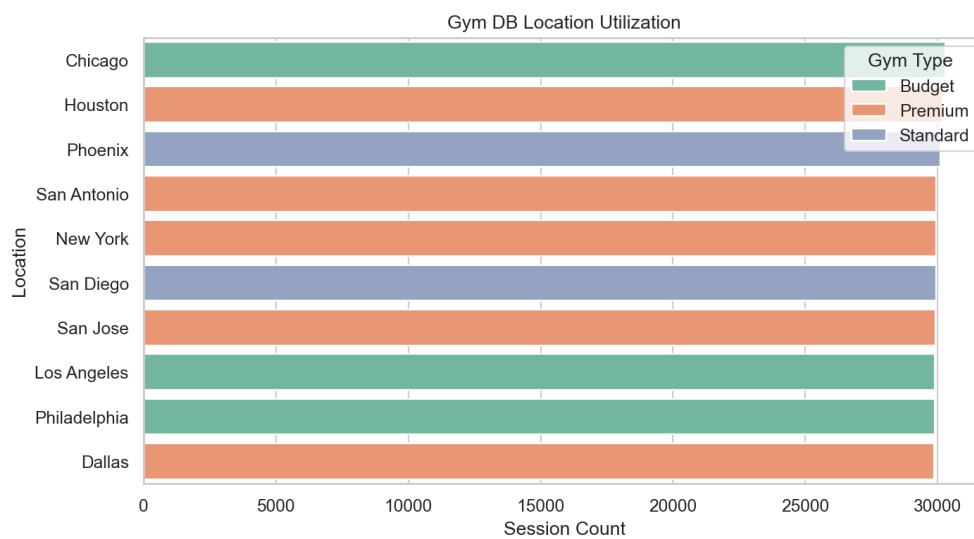


Figure 9: EDA: location utilization distribution from the Gym DB warehouse.

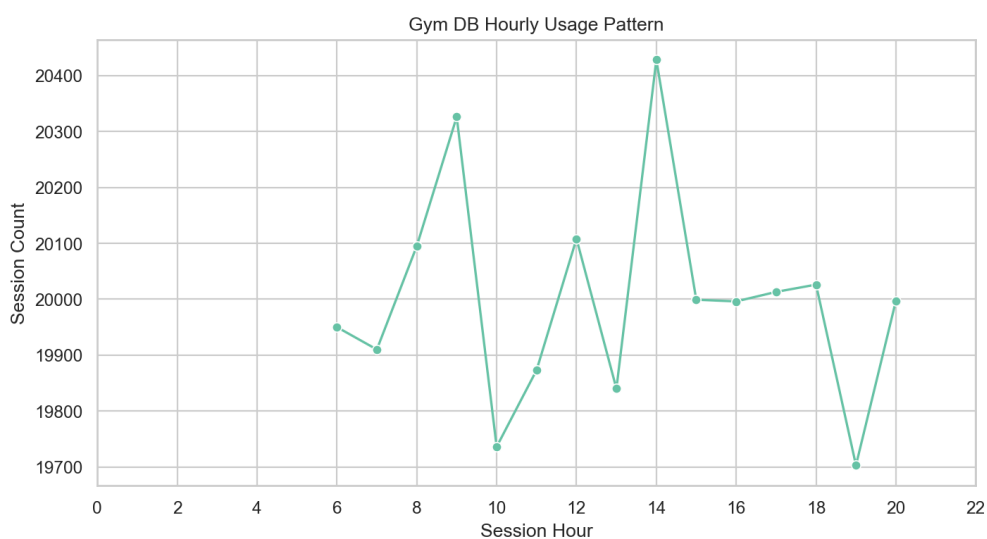


Figure 10: EDA: hourly usage pattern supporting staffing and schedule decisions.

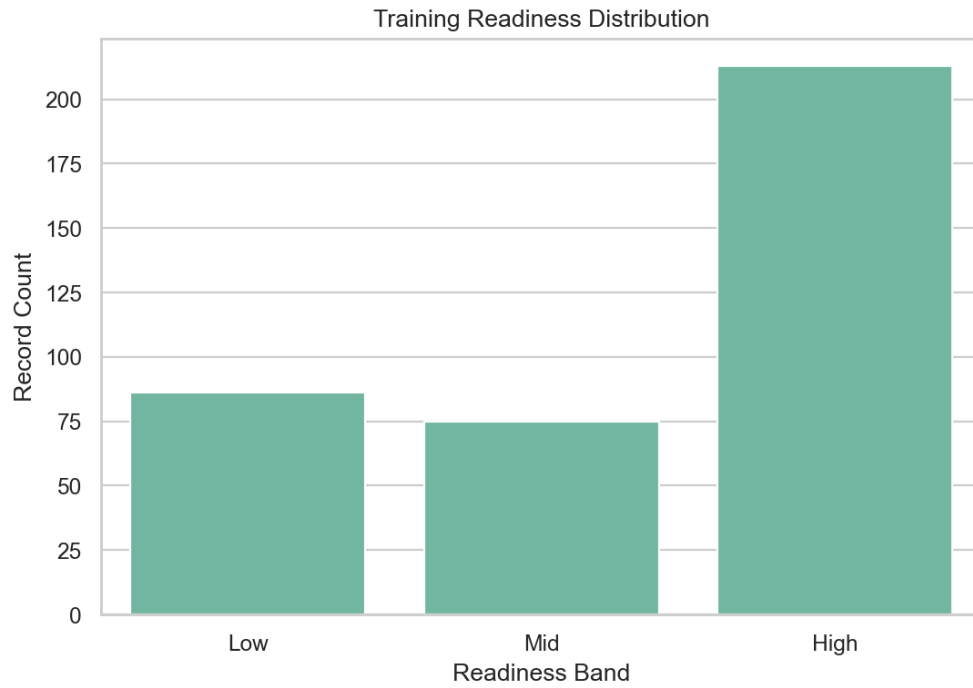


Figure 11: EDA: training readiness distribution from lifestyle-derived rules.

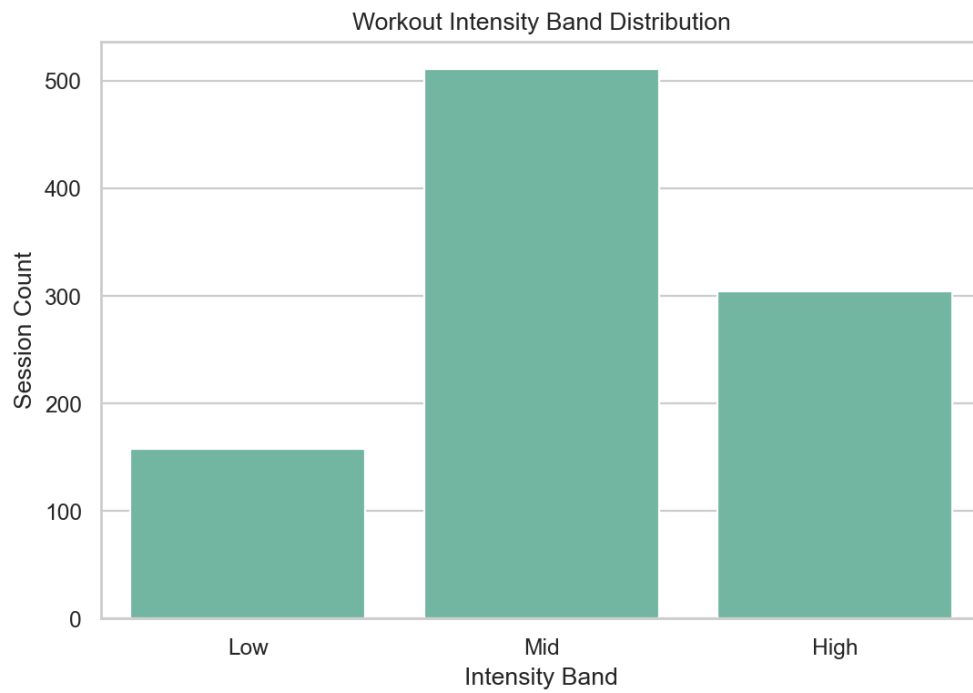


Figure 12: EDA: workout-intensity class distribution used to explain classifier imbalance.

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APPENDIX A

SOURCE CODE AVAILABILITY

The complete source code for Smart Workout is available at:

`https://github.com/your-repository`

A live deployment is accessible at:

`https://your-demo-url.com`

The repository includes:

- Source code
- Backend API
- Frontend
- Documentation